Differentiation — Product Rule Skill Ladder

ANSWER KEY

Chunk 0 — Spot the two factors

1. $u=x^2$, $v=e^x$
2. $u=x$, $v=\ln x$
3. $u=\sin x$, $v=\cos x$
4. $u=x^4$, $v=\tan x$
5. $u=e^x$, $v=\cos x$
6. $u=(x+1)$, $v=(x-3)$
7. $u=x$, $v=e^x$
8. $u=\ln x$, $v=\sin x$
9. $u=x^2$, $v=(x^3-2)$
10. $u=\cos x$, $v=x^5$

Chunk 1 — Differentiate each factor alone

1. $2x$
2. e^x
3. 1
4. $1/x$
5. $\cos x$
6. $-\sin x$
7. $4x^3$
8. $\sec^2 x$
9. 1
10. 1

Chunk 2 — Plug given pieces into the formula

1. **$dy/dx = 3x^2e^x + x^3e^x$**

   $(= x^2e^x(3+x))$

2. **$dy/dx = \ln x + 1$**

   $\ln x + x\cdot(1/x)$

3. **$dy/dx = \cos^2 x - \sin^2 x$**

   $\cos x\cdot\cos x + \sin x\cdot(-\sin x)$

4. **$dy/dx = 4x^3\tan x + x^4\sec^2 x$**

5. **$dy/dx = e^x\cos x - e^x\sin x$**

   $(= e^x(\cos x - \sin x))$

6. **$dy/dx = 2x - 2$**

   $(x-3)+(x+1)$

7. **$dy/dx = e^x(1+x)$**

   $e^x + x\cdot e^x$

8. **$dy/dx = \sin x/x + \ln x\cdot\cos x$**

9. **$dy/dx = 5x^4 - 4x$**

   $2x(x^3-2)+x^2(3x^2) = 2x^4-4x+3x^4$

10. **$dy/dx = 5x^4\cos x - x^5\sin x$**

Chunk 3 — Full product rule, one-term factors

1. **$dy/dx = 3x^2\sin x + x^3\cos x$**

2. **$dy/dx = e^x(1+x)$**

   $e^x + xe^x$

3. **$dy/dx = 2x\ln x + x$**

   $2x\cdot\ln x + x^2\cdot(1/x)$

4. **$dy/dx = e^x(\cos x - \sin x)$**

5. **$dy/dx = 4x^3\cos x - x^4\sin x$**

6. **$dy/dx = \ln x\cdot\cos x + \sin x/x$**

7. **$dy/dx = \tan x + x\cdot\sec^2 x$**

8. **$dy/dx = e^x(\sin x+\cos x)$**

9. **$dy/dx = 2x\cos x - x^2\sin x$**

10. **$dy/dx = e^x(1/x + \ln x)$**

    $(1/x)e^x + \ln x\cdot e^x$

Chunk 4 — Full product rule, multi-term factors

1. $dy/dx = 3x^2+2x-3$
2. $dy/dx = 6x^2-6x-5$
3. $dy/dx = 9x^2-2x+6$
4. $dy/dx = 4x^3+12x^2-1$
5. $dy/dx = 3x^2+2x-2$
6. $dy/dx = -3x^2+10x-3$
7. $dy/dx = 3x^2+4x-8$
8. $dy/dx = 9x^2+4x-15$
9. $dy/dx = 3x^2-12x-6$
10. $dy/dx = 6x^2+2x-18$

Chunk 5 — Full product rule, mixed function types

1. **$dy/dx = e^x(\cos x - \sin x)$**

2. **$dy/dx = 3x^2\tan x + x^3\sec^2 x$**

3. **$dy/dx = \cos x/x - \ln x \cdot \sin x$**

4. **$dy/dx = e^x(\cos x + \sin x)$**

5. **$dy/dx = x^4(5\ln x + 1)$**
   $5x^4\ln x + x^4$

6. **$dy/dx = \cos x$**
   Check: $y = \tan x \cdot \cos x$ simplifies to $\sin x$ first ($\tan x = \sin x/\cos x$), so $dy/dx = \cos x$ directly — matches the product-rule result.

7. **$dy/dx = e^x(\ln x + 1/x)$**

8. **$dy/dx = x^2\cos x + 2x \cdot \sin x$**

9. **$dy/dx = \cos x$**
   Same reasoning as Q6, reversed order — $y = \cos x \cdot \tan x = \sin x$, so $dy/dx = \cos x$.

10. **$dy/dx = x^3(1 + 4\ln x)$**
    $x^3 + 4x^3\ln x$